

Reduced-Belt Transient Mechanics

Final study context and results-writing reference

Basis: final artifact run **2026-09-16**, including broad/controlled trajectories, 28-case Baja envelope, equation sensitivity, high-contact/mixed-slip closure tests, controlled overrun, and belt-inertia continuation.

One-sentence result

For the investigated Baja-class CVT, belt-transport acceleration is a persistent first-order part of the reduced tension closure; shift-acceleration inertia becomes first-order during sufficiently rapid shift deceleration/reversal; path-curvature and moving-radius terms remain practically negligible across the tested envelope; and a coherent low-belt-inertia continuation stays within sub-percent state error without changing hybrid transition counts.

Study status: discovery complete for the stated Baja scope. Recommended next step is results-section synthesis, not another generic mechanics sweep.

This document is a frozen context note, not a manuscript draft. It records what the simulations support, why each conclusion is mechanically interpretable, and where the claim boundary remains.

1. Governing equations and where the studied terms come from

There are two separate belt balances. The whole-belt transport equation governs the single global belt transport speed:

$$m_b * v_{b_dot} + \tau_p / r_{p,eff} + \tau_s / r_{s,eff} = 0$$

The independent closed tension-loop compatibility can be organized as:

$$F_{shift_accel} + F_{path_curvature} + F_{belt_accel} + F_{moving_radius} + F_{contact} = 0$$

The four transient terms arise from the local wrap acceleration pieces $-r_j r_{j_ddot}$ (radial) and $r_j v_{b_dot} + r_{j_dot} v_b$ (tangential), after using $r_j = r_j(s)$:

Mechanism	Final-equation form	Physical driver
Shift acceleration	$-2 q (r_p r_{p'} - r_s r_{s'}) s_{ddot}$	s_{ddot} : radial acceleration from sheave acceleration
Path curvature	$-2 q (r_p r_{p''} - r_s r_{s''}) s_{dot}^2$	s_{dot}^2 : radial acceleration along nonlinear $r(s)$
Belt acceleration	$q (H_p r_p - H_s r_s) v_{b_dot}$	v_{b_dot} : belt material speeding up/slowing down
Moving radius	$q (H_p r_{p'} - H_s r_{s'}) s_{dot} v_b$	$s_{dot} v_b$: belt transport while wrap radii move
Net contact	$G_p N_p - G_s N_s$	difference of the two mapped contact-load contributions

Important interpretation: the gross contact pieces $G_p N_p$ and $G_s N_s$ can each be hundreds or thousands of newtons while nearly cancelling. A sub-newton transient term can therefore be important to the surviving compatibility equation even though it is tiny relative to gross clamp/contact force.

2. What was actually tested

The study intentionally progressed from broad discovery to targeted closure. This matters because the final conclusions are not based on one launch or one uphill event.

Stage	Coverage	Purpose
Broad + controlled	Flat launch/upshift; 8/18/28 deg load cases; +18 deg applied over 100 ms; 200-800 ms	Identify 200-800 ms hierarchy and transient-rate dependence
Baja envelope	28 cases; -20 to +30 deg; 50-800 ms; disturbance start across narrow range of belt speeds	Search for speed-Baja trajectory, load shift directions, ratios and distributions
Equation sensitivity	Full shift range; traction utilization to 0.95; independent primary/secondary coefficient growth	Expand coefficient growth beyond the exact simulated trajectory
High-contact closure	13 clamp-stress cases; 40/60/80/95% sticking targets; mixed slip	Test high-stipulation and contact asymmetry without changing friction
Controlled overrun	2 cases with downhill road plus primary resisting torque; power sign inverted	Test actual secondary-to-primary power flow, not merely unloading
Inertia continuation	Flat, fast backshift, fast unloading; row-only and coherent density distribution	Determine whether force-small whole-belt inertia is also trajectory

The 28-case Baja envelope covered pre-disturbance active shift fractions from 0.148061892548099 to 0.75 and belt speeds from 11.4 to 20.8 m/s. All 13 downhill/unloading cases drove further upshift; 9 of 15 uphill cases produced actual backshift.

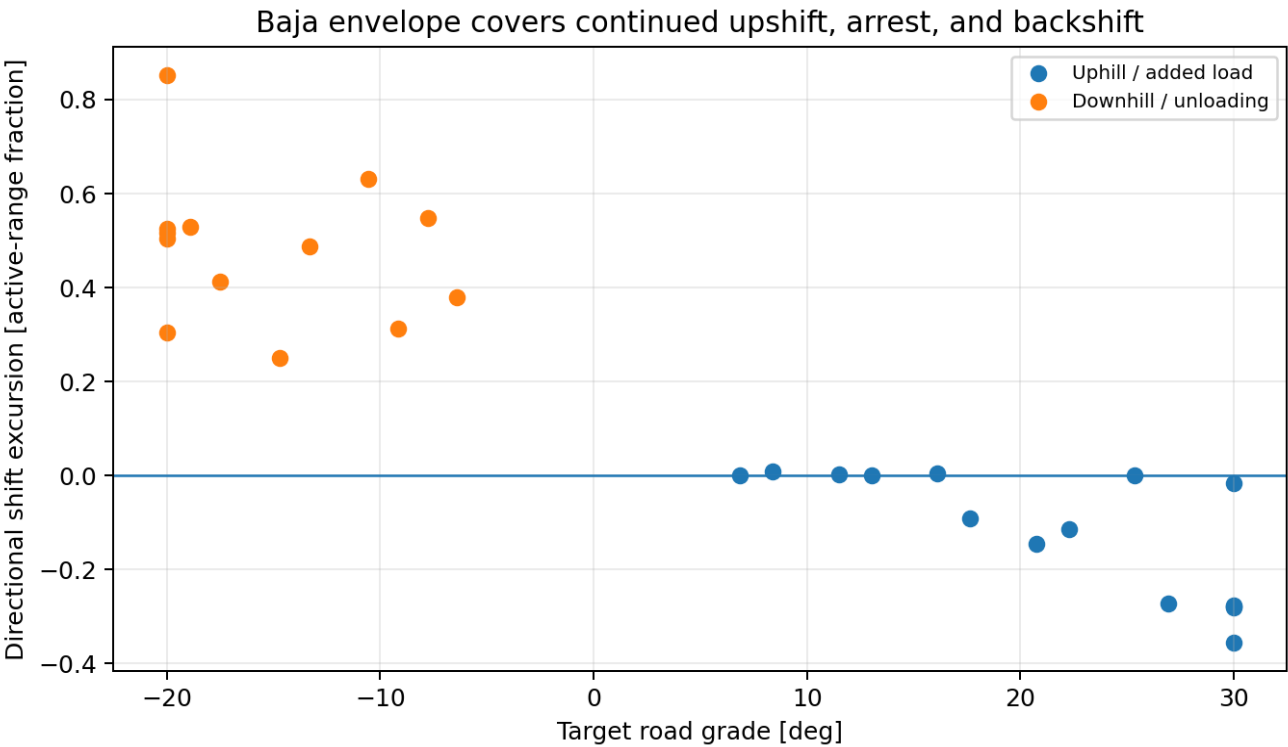


Figure 1. Directional response of the smart Baja envelope. Positive load does not automatically imply backshift; the set spans continued upshift, arrest, and true backshift.

3. Shift-acceleration inertia: transiently important, not generally dominant

The shift-acceleration term is proportional to s_{ddot} . The controlled +18 deg experiment separated disturbance magnitude from disturbance application time. The zero-rise case is an instantaneous road-load application - no CVT state is stepped.

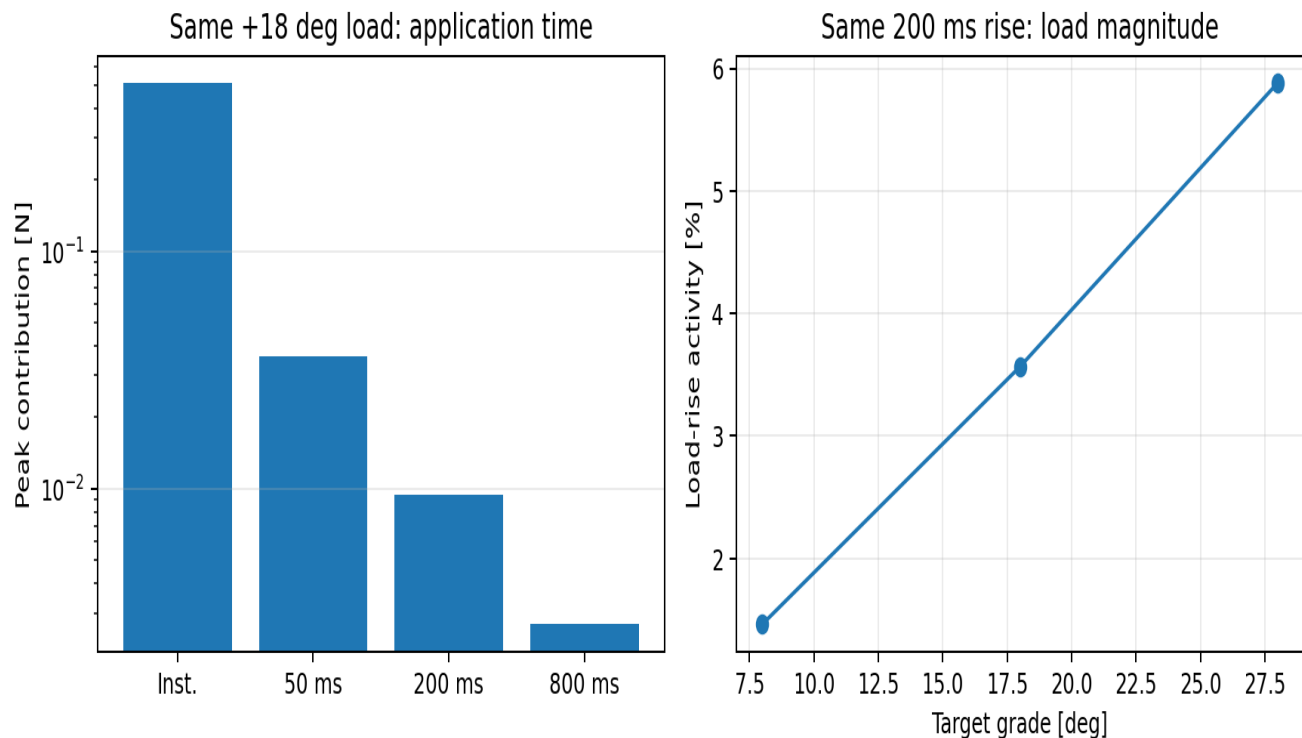


Figure 2. Controlled transient sensitivity. For the smooth 50/200/800 ms +18 deg cases, the fitted peak shift-acceleration contribution scales approximately as $t_{rise}^{-0.94}$, close to inverse rise time. At fixed 200 ms rise, stronger uphill load increases the term activity.

Across the Baja envelope, the mechanism becomes important when the actual sheave motion is strongly accelerated or reversed, not because "backshift" is intrinsically special.

4. Baja envelope connects the equation driver to realized importance

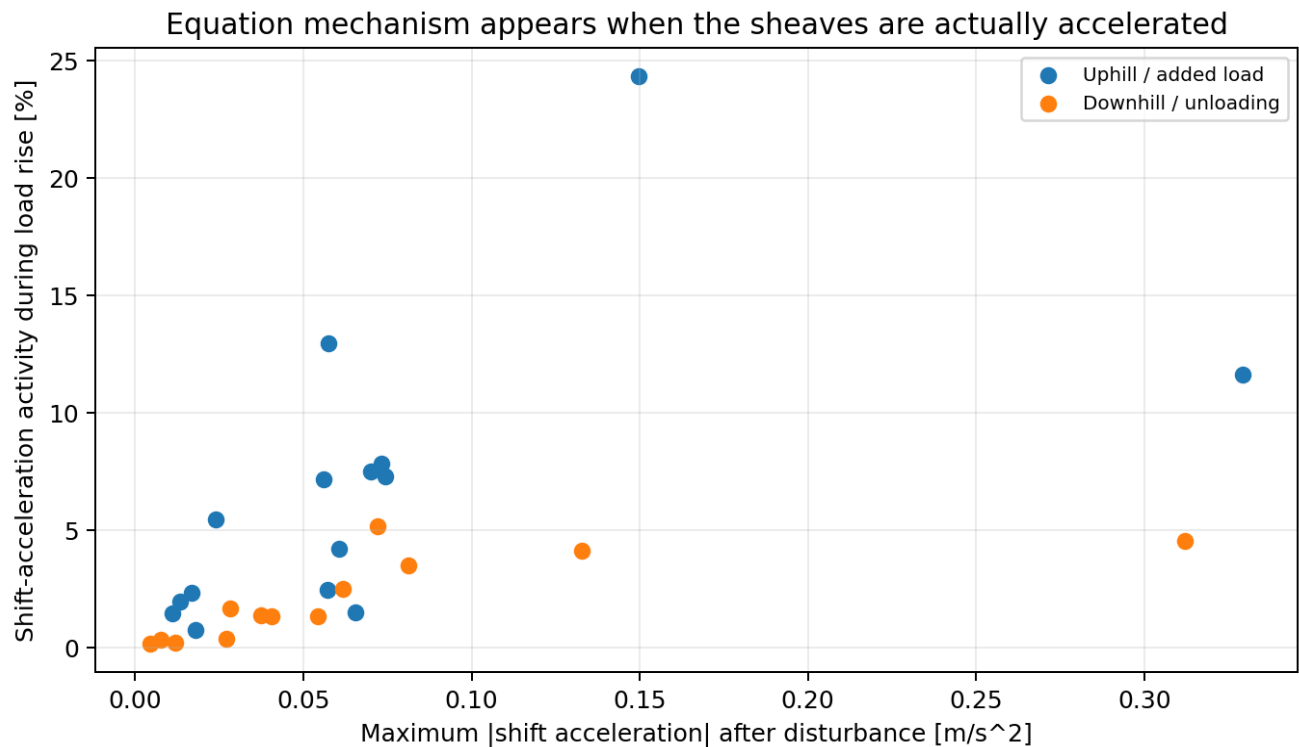


Figure 4. The broad envelope links the equation driver $|s_{ddot}|$ directly to the realized shift-acceleration activity during load application.

A representative fast uphill case (+20.7 deg applied over about 56 ms from ~64% shift) produced roughly 24% shift-acceleration activity during the load-rise window. This is the key evidence that the term can become first-order during rapid deceleration/reversal of shift motion.

In contrast, path-curvature activity stayed extremely small across ordinary upshift, backshift, unloading, overrun, high-contact stick, and slip tests. The coefficient itself remains finite; the limiting factor is the physically realized s_{dot}^2 driver.

5. Equation sensitivity explains what the Baja sweep did not reach

The original Baja envelope reached only about 19% of the static traction limit, so equation-level sensitivity was evaluated independently to 95% utilization and with primary/secondary utilization varied separately.

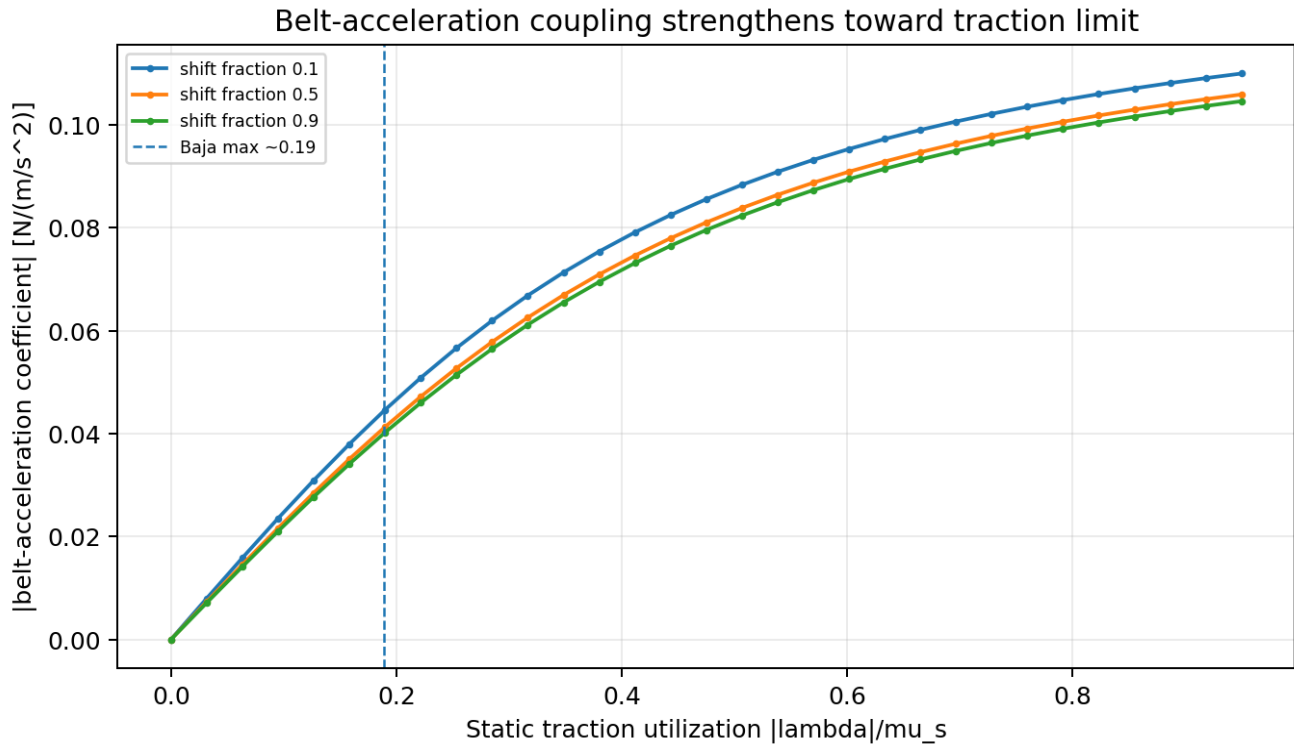


Figure 5. Belt-acceleration coupling strengthens materially as contact demand approaches the traction limit. This reinforces, rather than weakens, the conclusion that belt-transport acceleration should be retained.

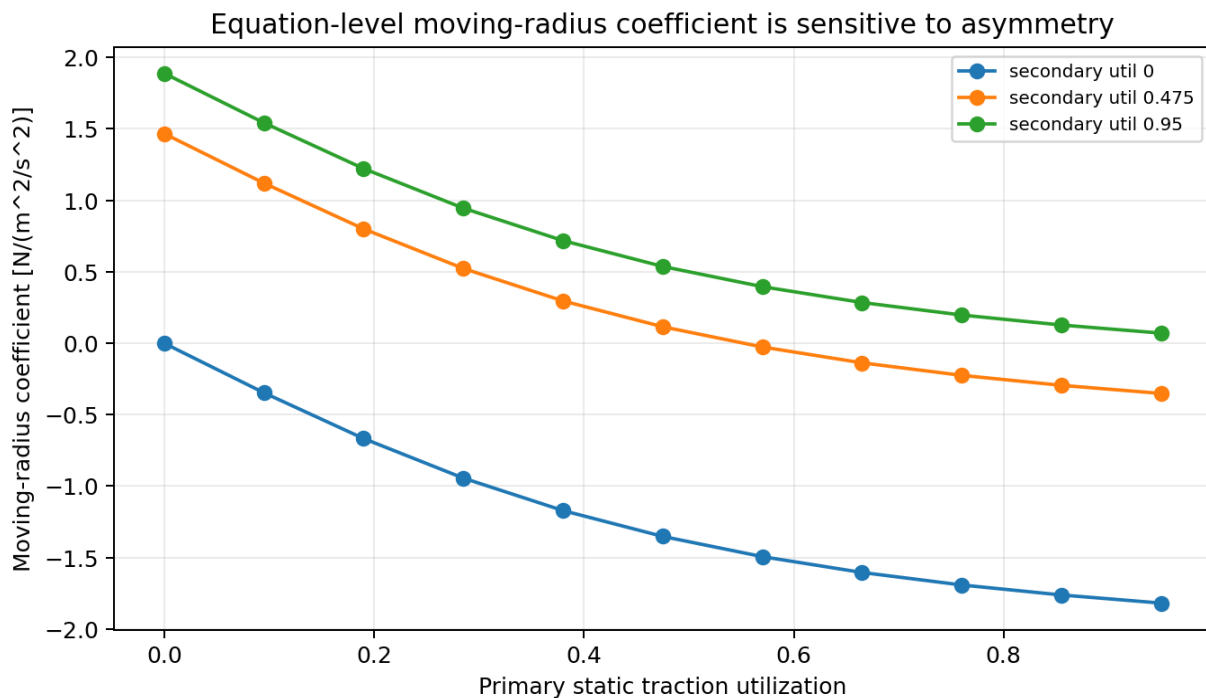


Figure 6. The moving-radius coefficient can become large when primary and secondary traction states are both high and asymmetric. This identified the final unresolved physical regime to test.

6. High-contact and mixed-slip closure

Friction coefficients were held fixed. Contact demand was raised by weakening clamp mechanisms, letting CINDER decide when stick failed. The study reached sticking states near 40%, 60%, 80%, and 95% utilization, then naturally entered primary-slip/secondary-stick; the most aggressive primary-clamp case also reached both-slip.

Target stick utilization	Realized	Case	Shift speed	Belt speed
40%	40.0%	contact_primary_45	0.000 m/s	13.28 m/s
60%	60.1%	contact_primary_45	0.000 m/s	12.70 m/s
80%	80.4%	contact_combined_p40_s20	0.000 m/s	12.81 m/s
95%	95.1%	contact_combined_p40_s20	0.000 m/s	12.56 m/s

The maximum realized moving-radius coefficient reached $0.928 \text{ N}/(\text{m}^2/\text{s}^2)$, validating the equation-level prediction that contact asymmetry can amplify the coefficient. Yet the largest realized moving-radius force was only 0.0182 N , and the maximum instantaneous activity over these stress cases was only 0.130% .

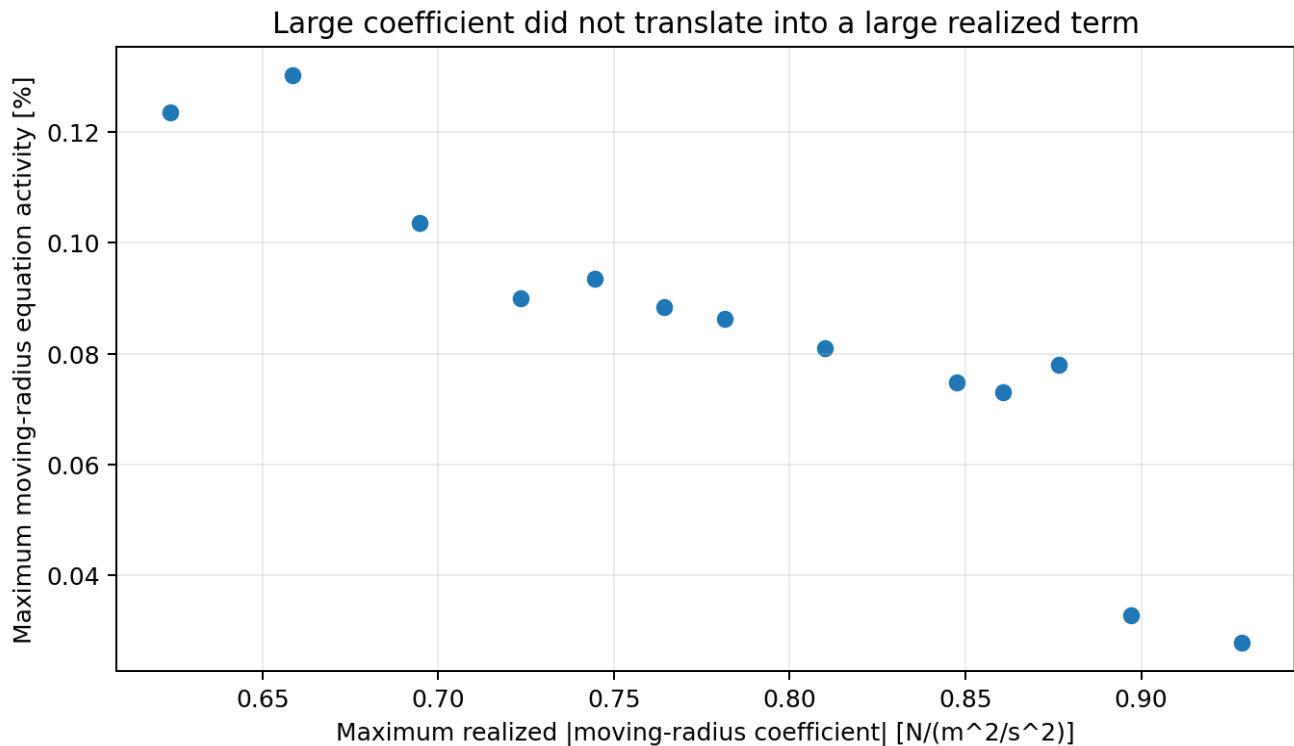


Figure 7. High/asymmetric contact makes the coefficient large, but the actual moving-radius contribution remains tiny because the full driver $s_{\text{dot}} * v_b$ is not simultaneously large.

7. Mixed slip: the coefficient grows, but the driver still suppresses the term

The slipping interface uses the kinetic traction limit while the sticking interface is solved inside its static bound. The realized mixed branch was primary-slip/secondary-stick; both-slip also occurred once. Primary-stick/secondary-slip did not arise naturally despite aggressive secondary-reaction weakening.

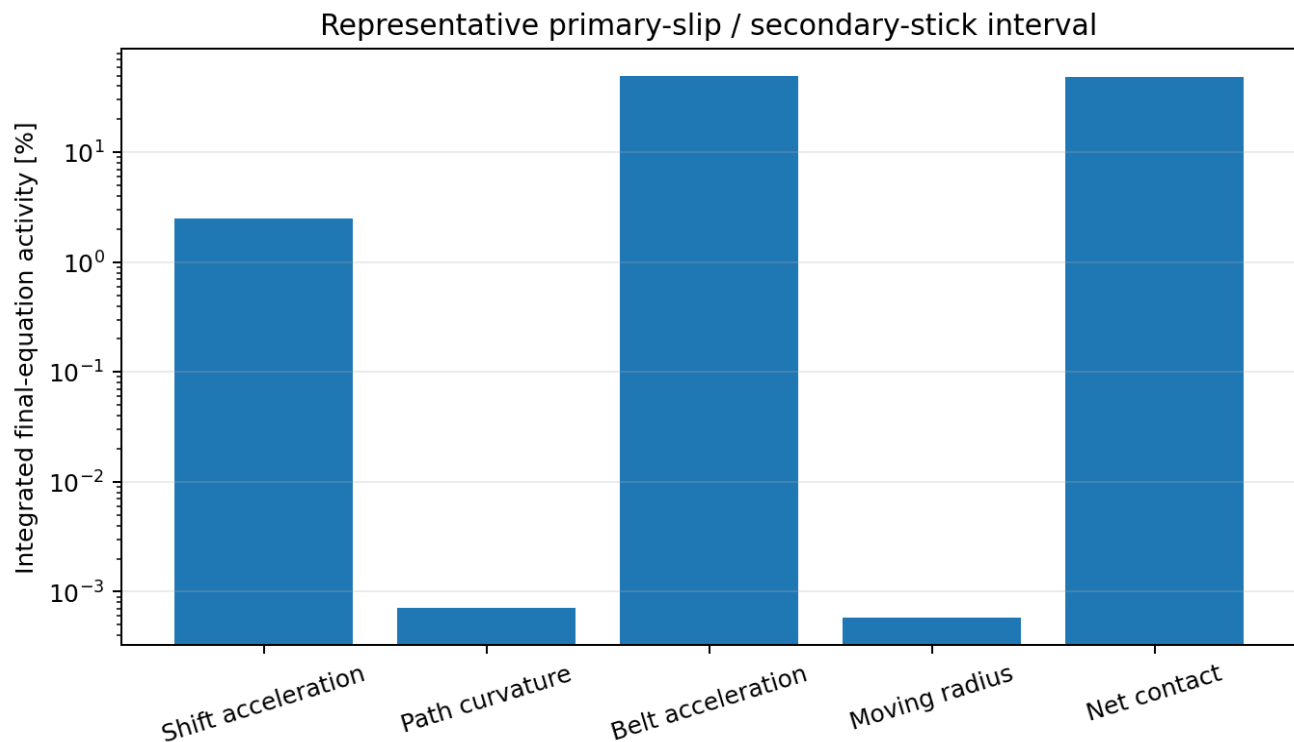


Figure 8. Representative active primary-slip/secondary-stick interval. Belt acceleration and net contact still dominate; moving-radius activity is orders of magnitude smaller.

Across mixed-slip examples, the largest moving-radius contribution was 0.0182 N. The strongest recorded mixed-slip moving-radius activity was 0.124% at an instant, while an integrated representative mixed-slip interval was only about 0.0584%.

Physical explanation: the large contact-asymmetry coefficient tends to occur either near engagement, where belt speed is still small, or in high-utilization seated states where shift speed is near zero. The product $\dot{s} * v_b$ therefore remains small.

8. True overrun confirms the direction-independent transient picture

The earlier downhill envelope represented unloading under the normal full-throttle engine boundary. The final stage added primary resisting torque and verified actual secondary-to-primary power flow using power signs.

Case	True-overrun samples	Min primary power	Max secondary power
overrun_mild	80.7%	-1.72 kW	12.99 kW
overrun_strong	82.8%	-3.70 kW	11.88 kW

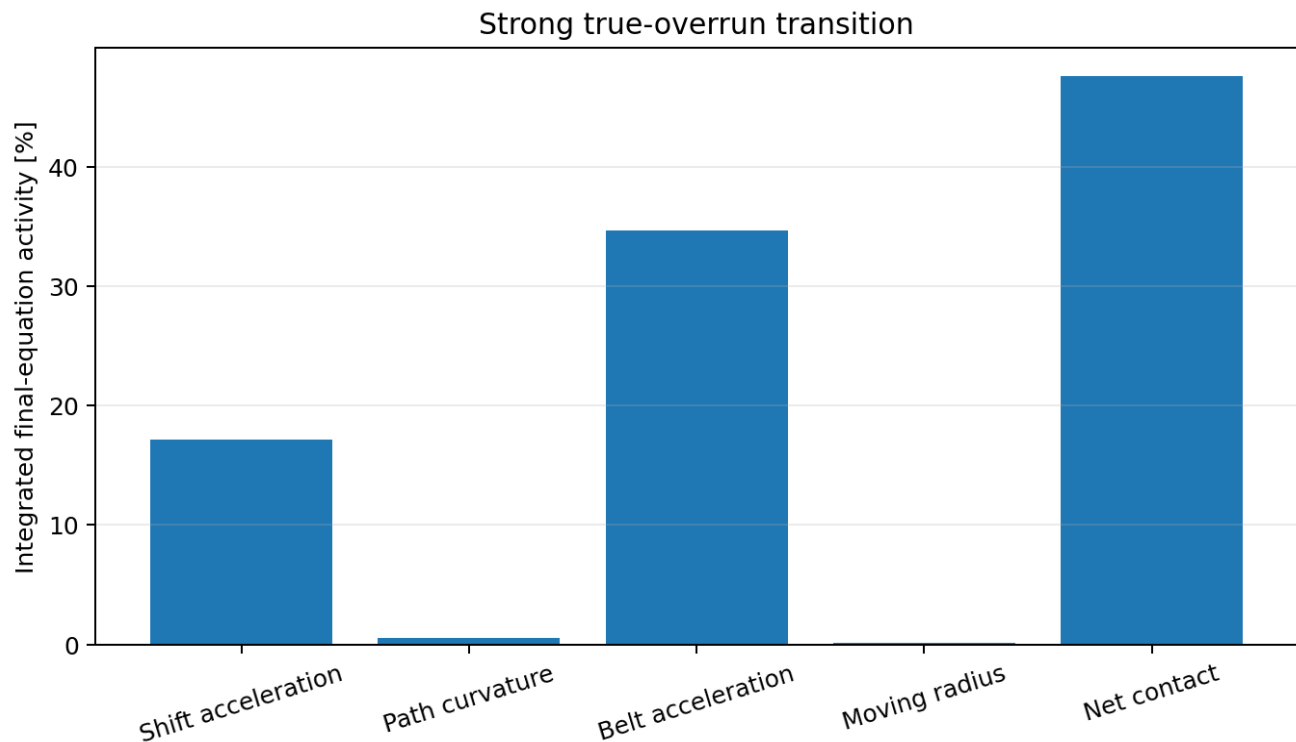


Figure 9. Strong overrun transition. Shift acceleration again becomes significant (~17% integrated activity), showing that the mechanism is tied to rapid change/reversal of shift motion rather than specifically to uphill backshift.

Moving-radius transport remains minor even under reversed power flow, while belt acceleration remains a major part of the compatibility balance.

9. Whole-belt inertia: force-small and close to a coherent low-inertia limit

The global transport equation is $m_b v_b \dot{+} \tau_p/r_p + \tau_s/r_s = 0$. Across the Baja envelope the whole-belt inertia term was typically far smaller than the two opposing pulley traction reactions. The final question was whether that force-small term is also trajectory-small.

Two continuations were tested. Scaling only the global transport-row coefficient to 10% or 3% failed at engagement in all three scenarios because no admissible contact branch remained. This is not evidence that physical belt inertia is essential; it shows that deleting one row term in isolation is structurally inconsistent with the rest of the retained belt inertia/contact mechanics.

The coherent continuation instead scaled belt density, making global belt mass and local wrap inertia vanish together. Every 10% and 3% density case completed with exactly the same hybrid transition count as its reference.

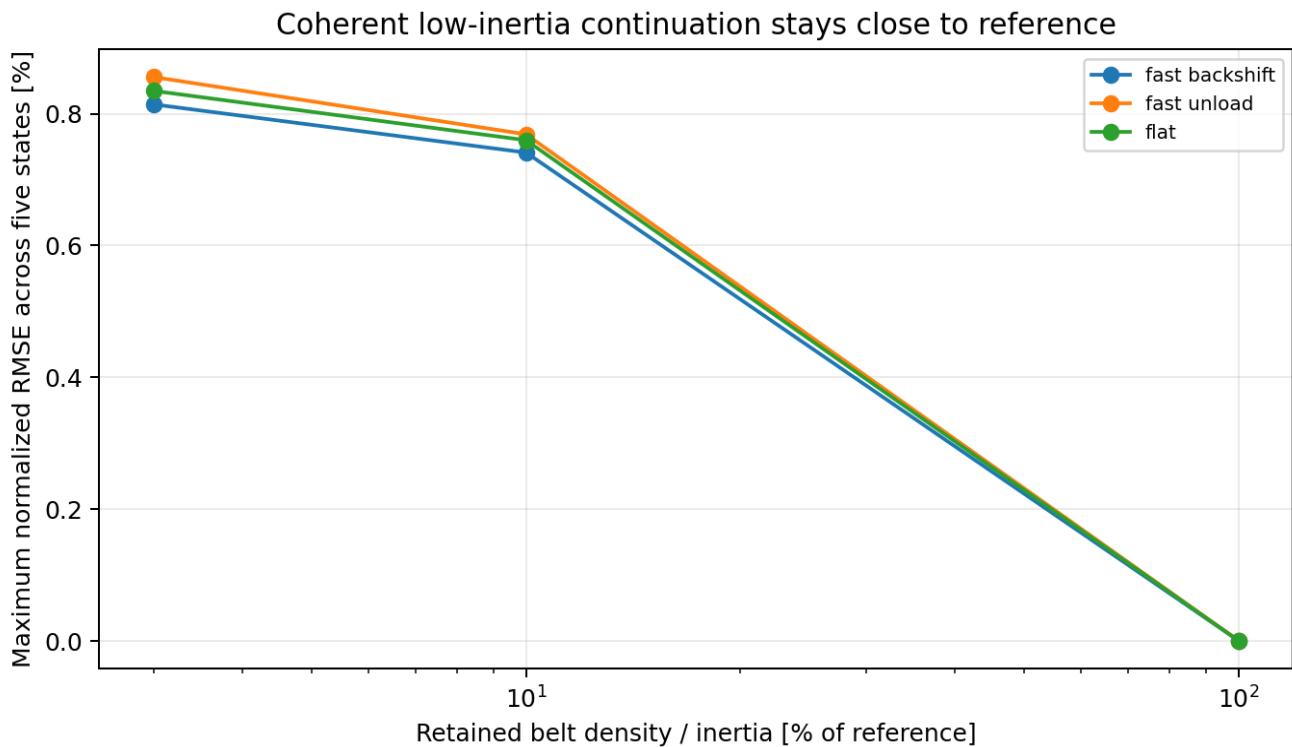


Figure 10. Coherent belt-density continuation. Even at 3% retained belt inertia, the worst normalized RMSE among the five states remains below about 0.86% across flat acceleration, fast backshift, and fast unloading.

Scenario	3% density: max state nRMSE	Transition count
flat	83.480%	15 (delta +0)
fast backshift	81.423%	15 (delta +0)
fast unload	85.581%	16 (delta +0)

10. Final term-by-term interpretation

Term	Evidence	Final interpretation for investigated Baja scope
Whole-belt inertia $m_b v_b \dot{v}_b$	Force-small across envelope; coherent density continuous	Small. $A \sim 0.86\%$ was sustained, $M \sim 0.1\%$, reducing by bridge
Shift-acceleration inertia $\sim s_{dot} \dot{v}_b$	Strong inverse-timescale response; 10-25% equation	Relatively important in smooth operation, first-order during rap
Path curvature $\sim s_{dot}^2$	Remained tiny across upshift, backshift, overrun, high	Negligible, tested Baja operation because realized sheave speeds are too low, no
Belt acceleration $\sim v_b \dot{v}_b$	Persistent ~first-order partner of net contact; coefficient	Clearly remains core partner of reduced tension compatibility.
Moving radius $\sim s_{dot} v_b$	Equation coefficient can approach ~1 under high/asym	Practically negligible, tested for Baja $\sim 0.01\%$. Coefficient sensitivity $\sim 0.1\%$ traj
Net contact $G_p N_p - G_s N_s$	Persistent dominant balancing contribution	Core closure term; gross contact pieces can be much larger than their surviving diff

Overall decision: the discovery phase is complete for this Baja-class application. Further generic sweeps are unlikely to change the main physical hierarchy. Future work should focus on deriving/evaluating any desired reduced formulation rather than searching for more exotic cases.

11. Claim boundary and caveats to preserve in the paper

What the study supports

The conclusions above are supported for the investigated reduced CINDER belt formulation and the Baja-class operating/tuning envelope sampled. The study intentionally spans both shift directions, disturbance timescales, ratio/speed range, high static utilization, mixed slip, both slip, true overrun, and plausible secondary-preload variation.

What it does not support

Do not overclaim	Why
"Moving-radius transport is mathematically negligible for every conceivable case."	The effect can be quite large outside the low-contact symmetric regime; the result is that physics is not captured.
"Every contact branch was dynamically sampled."	Primary-stick/secondary-slip was not naturally realized. Primary-slip/secondary-stick and both-slip were not sampled.
"Zero belt density was solved and proven equivalent."	The continuation reached 3% density, not an exact massless DAE limit. It strongly suggests a near-zero limit.
"Downhill envelope cases were engine braking."	The broad downhill cases were unloading under full-throttle drive; true overrun was separately created.
"Small relative to gross clamp force means dynamical irrelevance."	Final relevance importance must be evaluated after cancellation of gross contact contributions.

Unrealized branch note: primary-stick/secondary-slip did not occur despite aggressive secondary-reaction weakening. For the Baja results this is best reported as an observed branch preference/limitation, not forced artificially merely to complete a checklist.

12. Results-writing cheat sheet

Suggested results narrative order:

1. Start from the two final belt equations and state that only surviving terms were decomposed. 2. Establish ordinary free-stick hierarchy. 3. Show controlled timescale dependence of shift acceleration. 4. Expand to the smart Baja envelope and both shift directions. 5. Use equation sensitivity to identify high-contact/asymmetric moving-radius risk. 6. Show the targeted high-contact/mixed-slip experiment closes that risk. 7. Confirm overrun. 8. Finish with the coherent low-inertia continuation.

Useful quantitative anchors

Anchor	Value / observation
Controlled +18 deg load	Peak shift-accel term: 0.0360 N at 50 ms, 0.00941 N at 200 ms, 0.00269 N at 800 ms; smooth-case fit $\sim t_{\text{rise}}^{-0.94}$
Baja envelope	28 cases; pre-shift fraction 0.15-0.75; pre-belt speed 11.4-20.8 m/s; 9/15 uphill cases backshift; 13/13 downhill cases forward
Baja contact demand	Broad envelope max $\sim 18.9\%$ static utilization
High-contact stick	Targets near 40%, 60%, 80%, 95% reached without altering friction coefficients
Mixed slip	Primary-slip/secondary-stick repeatedly realized; both-slip realized once; opposite mixed branch not naturally realized
Moving-radius closure	Max coefficient $\sim 0.928 \text{ N}/(\text{m}^2/\text{s}^2)$, max realized force $\sim 0.0182 \text{ N}$, max stress-case instantaneous activity $\sim 0.130\%$
True overrun	$\sim 81\text{-}83\%$ of post-event samples satisfy reversed power flow; strong case primary absorbs $\sim 3.70 \text{ kW}$ while secondary slips
Coherent inertia continuation	At 3% belt density/inertia: worst five-state normalized RMSE 0.81-0.86%; transition counts unchanged in all three scenarios

Recommended closing sentence for the study logic (adapt as needed):

"The retained transient terms separate into mechanisms that are persistently first-order, mechanisms activated only by rapid shift acceleration, and mechanisms whose equation coefficients may be finite or even large but whose complete physical drivers remain too small over the investigated Baja operating envelope to materially affect the closed tension balance."

Key source files in the frozen artifact package: controlled_load_summary.csv; baja_envelope_observed_coverage.csv; equation_sensitivity/*.csv; contact_closure_summary.csv; contact_stick_targets.csv; mixed_slip_examples.csv; overrun_summary.csv; inertia_continuation_comparison.csv; closure_failures.csv; stage_status.json.